CST476	MOBILE COMPUTING	CATEGORY	L	T	P	CREDIT	YEAR OF INTRODUCTION
		PEC	2	1	0	3	2019

Preamble: The course is designed with the view of preparing the engineering students capable of understanding the communication protocols, various architectures and security features used in mobile computing. This course covers basics of mobile computing, architecture of wireless transmission systems and next generation networks. This course enables the learners to acquire advanced concepts on wireless communication systems and mobile ad-hoc networks.

Prerequisite: A sound knowledge of computer networks.

Course Outcomes: After the completion of the course the student will be able to

CO#	Course Outcomes
CO1	Explain the various mobile computing applications, services, design considerations and architectures (**Cognitive knowledge: Understand**)
CO2	Describe the various technology trends for next generation cellular wireless networks and use the spreading concept on data transmission **(Cognitive knowledge: Apply)**
CO3	Summarize the architecture of various wireless LAN technologies **(Cognitive knowledge: Understand)**
CO4	Identify the functionalities of mobile network layer and transport layer **(Cognitive knowledge: Understand)**
CO5	Explain the features of Wireless Application Protocol **(Cognitive knowledge: Understand)**
CO6	Interpret the security issues in mobile computing and next generation technologies **(Cognitive knowledge: Understand)**

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✔	✔										✔
CO2	✔	✔	✔	✔								✔
CO3	✔	✔	✔									✔
CO4	✔	✔	✔		✔							✔

CO5	✓	✓	✓									✓
CO6	✓	✓	✓									✓

Abstract POs defined by National Board of Accreditation			
PO#	**Broad PO**	**PO#**	**Broad PO**
PO1	Engineering Knowledge	**PO7**	Environment and Sustainability
PO2	Problem Analysis	**PO8**	Ethics
PO3	Design/Development of solutions	**PO9**	Individual and team work
PO4	Conduct investigations of complex problems	**PO10**	Communication
PO5	Modern tool usage	**PO11**	Project Management and Finance
PO6	The Engineer and Society	**PO12**	Life long learning

Assessment Pattern

Bloom's Category	Test 1 (%)	Test 2 (%)	End Semester Examination (%)
Remember	30	30	30
Understand	50	50	50
Apply	20	20	20
Analyse			
Evaluate			
Create			

Mark Distribution

Total Marks	CIE Marks	ESE Marks	ESE Duration
150	50	100	3

Continuous Internal Evaluation Pattern:

Attendance	: **10 marks**
Continuous Assessment Test	: **25 marks**
Continuous Assessment Assignment	: **15 marks**

Internal Examination Pattern:

Each of the two internal examinations has to be conducted out of 50 marks. First series test shall be preferably conducted after completing the first half of the syllabus and the second series test shall be preferably conducted after completing remaining part of the syllabus. There will be two parts: Part A and Part B. Part A contains 5 questions (preferably, 2 questions each from the completed modules and 1 question from the partly completed module), having 3 marks for each question adding up to 15 marks for part A. Students should answer all questions from Part A. Part B contains 7 questions (preferably, 3 questions each from the completed modules and 1 question from the partly completed module), each with 7 marks. Out of the 7 questions, a student should answer any 5.

End Semester Examination Pattern:

There will be two parts; Part A and Part B. Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 questions from each module of which a student should answer any one. Each question can have maximum 2 sub-divisions and carries 14 marks.

Syllabus

Module - 1 (Mobile Computing Basics)

Introduction to mobile computing – Functions, Middleware and Gateways, Application and services. Mobile computing architecture – Internet: The Ubiquitous network, Three-tier architecture for Mobile Computing, Design considerations for mobile computing.

Module – 2 (Wireless Transmission and Communication Systems)

Spread spectrum – Direct sequence, Frequency hopping. Medium Access Control – Space Division Multiple Access (SDMA), Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), Code Division Multiple Access (CDMA). Satellite Systems – Basics, Applications, Geostationary Earth Orbit (GEO), Low Earth Orbit (LEO), Medium Earth Orbit (MEO), Routing, Localization, Handover. Telecommunication Systems - Global System for Mobile Communication (GSM) services, Architecture, Handover, Security.

Module – 3 (Wireless LANs)

Wireless LAN - Advantages, Design goals, Applications, Infrastructure Vs Ad-hoc mode, IEEE 802.11 System Architecture, Protocol Architecture, Physical layer, Medium Access Control layer, HIPERLAN-1, Bluetooth.

Module – 4 (Mobile Network and Transport Layer)

Mobile network layer – Mobile Internet Protocol (IP), Dynamic Host Configuration Protocol (DHCP), Mobile ad-hoc networks – Routing, Dynamic Source Routing (DSR), Destination Sequence Distance Vector (DSDV), Ad-hoc routing protocols. Mobile transport layer – Traditional Transmission Control Protocol (TCP), Improvements in Classical TCP. Wireless Application Protocol (WAP) - Architecture, Wireless Datagram Protocol (WDP), Wireless Transport Layer Security (WTLS), Wireless Transaction Protocol (WTP), Wireless Session Protocol (WSP).

Module – 5 (Mobile Security and Next Generation Networks)

Security issues in mobile computing - Information security, Security techniques and algorithms, Security models. Next generation networks - Orthogonal Frequency Division Multiplexing (OFDM), Wireless Asynchronous Transfer Mode (WATM), Multi Protocol Label Switching (MPLS), 10 pillars of 5G, Security for 5G communication.

Text Books

1. Asoke K. Talukder, Hasan Ahmad, Roopa R Yavagal, Mobile Computing Technology-Application and Service Creation, 2/e, McGraw Hill Education.
2. Jochen Schiller, Mobile Communications, Pearson Education Asia, 2008.
3. Jonathan Rodriguez, Fundamentals of 5G Mobile Networks, Wiley Publishers, 2015.

Reference Books

1. Raj Kamal, Mobile Computing, 2/e, Oxford University Press.
2. Andrew S. Tanenbaum, Computer Networks, PHI, 3/e, 2003
3. Theodore S. Rappaport, Wireless Communications Principles and Practice, 2/e, PHI, New Delhi, 2004.
4. Curt M. White, Fundamentals of Networking and Communication 7/e, Cengage learning.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Give examples for five mobile computing applications.
2. Identify any three differences between middleware and gateways.

Course Outcome 2 (CO2):

1. There are four stations sending data 1,1,1,0 respectively. Station 3 receives station 1's

data. Show the encoding, decoding and channel sharing mechanisms using CDMA.
2. Compare the influence of near/far effect and its countermeasures in TDMA and CDMA systems.

Course Outcome 3 (CO3):

1. Compare IEEE 802.11 and Bluetooth with respect to their ad-hoc capabilities.
2. Describe with neat sketch the major baseband states of a Bluetooth device.

Course Outcome 4 (CO4):

1. With the help of an example, show how routing process is handled by Dynamic Source Routing protocol.
2. Describe the major differences between AODV and the standard Distance Vector Routing algorithm. Why are extensions needed?
3. Simulate routing protocols using NS2.

Course Outcome 5 (CO5):

1. How does WAP push operation differ from pull operation?
2. With the help of a neat sketch explain the secure session establishment using WTLS.

Course Outcome 6 (CO6):

1. Explain the 3GPP security framework for mobile security.
2. Explain the features of policy-based security model.

Model Question Paper

QP CODE: **PAGES: 3**

Reg No:_______________
Name:_________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
EIGHTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: CST476
Course Name : Mobile Computing

Max Marks: 100 **Duration: 3 Hours**

PART-A
(Answer All Questions. Each question carries 3 marks)

1. Explain the different types of middleware and gateways in the architecture of mobile computing.

2. Explain the major segments to support mobile computing functions.

3. Compare and contrast the satellite systems – GEO, LEO and MEO.
4. Assume all stations can hear all other stations. One station wants to transmit and senses the carrier idle. Why can a collision still occur after the start of transmission?
5. List any three advantages and disadvantages of wireless LANs.
6. Compare the features of infrastructure and ad-hoc networks.
7. Mention the basic purpose of DHCP. Also list the entities of DHCP.
8. Identify the benefits of location information for routing in ad-hoc networks.
9. List any six pillars of 5G.
10. How does multifactor security model provide security in a mobile network?

(10x3=30)

Part B

(Answer any one question from each module. Each question carries 14 Marks)

11. (a) Describe the design considerations of three tier architecture of mobile computing. **(6)**

(b) Explain any four functions and applications of mobile computing. **(8)**

OR

12. (a) Explain Internet-Ubiquitous networks mentioning the significance and functions of core, edge and access network. **(6)**

(b) With the help of a neat sketch explain the three-tier architecture of mobile computing. **(8)**

13. (a) Check to see if the following set of chips can belong to an orthogonal system. **(6)**

[+1, +1, +1, +1] , [+1, -1, -1, +1] , [-1, +1, +1, -1] , [+1, -1, -1, +1]

(b) Summarize the routing and localization process in satellite systems. **(8)**

OR

14. (a) Apply Direct Sequence Spread Spectrum to the data 101 using Barker sequence 10110111000. Show the encoding and decoding steps. **(6)**

(b) Describe the system architecture of GSM networks. **(8)**

15. (a) How is Quality-of-Service provided in Bluetooth? **(6)**

(b) Explain the phases in Elimination-yield non-preemptive priority multiple access of HIPERLAN-1. **(8)**

OR

16. (a) Describe the protocol architecture of IEEE 802.11. **(6)**

(b) Explain the Medium Access Control management features provided in an IEEE 802.11 station. **(8)**

17. (a) With the help of an example, show the routing table creation using Destination Sequence Distance Vector Routing protocol in mobile ad-hoc networks. **(7)**

(b) Describe the router discovery methods used in mobile IP. **(7)**

OR

18. (a) Compare the features of flat routing and hierarchical routing. **(6)**

(b) List the entities of a mobile IP. With the help of an example, explain how packet delivery is done to and from a fixed node. **(8)**

19. (a) How is orthogonality helpful in Orthogonal Frequency Division Multiplexing? **(4)**

(b) Explain the functioning of Multi Protocol Label Switching technology. **(10)**

OR

20. (a) Describe the services of Wireless Asynchronous Transfer Mode. **(6)**

(b) Explain the different security models in mobile computing. **(8)**

TEACHING PLAN

No	Contents	No.of Lecture Hrs (35 hrs)
Module – 1 (Mobile Computing Basics) (6 hrs)		
1.1	Introduction to mobile computing – Functions	1
1.2	Middleware and Gateways	1
1.3	Application and services	1
1.4	Internet: The Ubiquitous network	1

1.5	Three-tier architecture for Mobile Computing	1
1.6	Design considerations for mobile computing	1
Module – 2 (Wireless Transmission and Communication Systems) (8 hrs)		
2.1	Direct sequence spread spectrum, Frequency hopping spread spectrum	1
2.2	Space Division Multiple Access (SDMA), Frequency Division Multiple Access (FDMA)	1
2.3	Time Division Multiple Access (TDMA)	1
2.4	Code Division Multiple Access (CDMA)	1
2.5	Satellite Systems Basics, Applications, Geostationary Earth Orbit (GEO), Low Earth Orbit (LEO), Medium Earth Orbit (MEO)	1
2.6	Routing, Localization, Handover	1
2.7	Global System for Mobile Communication (GSM) services, Architecture	1
2.8	Handover, Security	1
Module - 3 (Wireless LANs) (7 hrs)		
3.1	Wireless LAN - Advantages, Design goals, Applications, Infrastructure Vs Ad-hoc mode	1
3.2	IEEE 802.11 System Architecture	1
3.3	Protocol Architecture	1
3.4	Physical layer	1
3.5	Medium Access Control layer	1
3.6	HIPERLAN-1	1
3.7	Bluetooth	1
Module - 4 (Mobile Network and Transport Layer) (8 hrs)		
4.1	Mobile Internet Protocol (IP), Dynamic Host Configuration Protocol (DHCP)	1
4.2	Mobile ad-hoc networks – Routing, Dynamic Source Routing (DSR)	1

4.3	Destination Sequence Distance Vector (DSDV)	1
4.4	Ad-hoc routing protocols	1
4.5	Traditional Transmission Control Protocol (TCP), Improvements in Classical TCP	1
4.6	Wireless Application Protocol (WAP) – Architecture, Wireless Datagram Protocol (WDP)	1
4.7	Wireless Transport Layer Security (WTLS)	1
4.8	Wireless Transaction Protocol (WTP), Wireless Session Protocol (WSP)	1
Module - 5 (Mobile Security and Next Generation Networks) (6 hrs)		
5.1	Information security, Security techniques	1
5.2	Security algorithms, Security models	1
5.3	Introduction to Next generation networks, Orthogonal Frequency Division Multiplexing (OFDM)	1
5.4	Wireless Asynchronous Transfer Mode (WATM)	1
5.5	Multi Protocol Label Switching (MPLS)	1
5.6	10 pillars of 5G, Security for 5G communication	1